

Weathering the Vote: How Temperature Extremes Shape Political Participation in India

Shweta Bhogale[✉] A. Patrick Behrer[✉] Priyadarshi Amar[✉]

Bhavya Srivastava[✉] Anwesha Bhattacharya[✉]

January 8, 2026

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Abstract

We examine the causal impact of polling day temperature on voter turnout and electoral outcomes using comprehensive data from Indian state assembly elections. Exploiting plausibly exogenous variation in temperature deviations from historical norms on every polling day across elections from 1977 to 2023, we find that higher temperatures significantly increase voter turnout, with strongly non-linear effects. Temperature deviations of polling day temperatures exceeding thirty-year average temperatures by more than 2°C increase aggregate turnout by 0.8 percentage points. These effects exhibit important heterogeneity: female turnout responds more strongly to temperature variations than male turnout (0.83 vs 0.67 percentage points), and responses vary dramatically by long-term climatic conditions, highlighting important adaptation margins. In the coolest regions, warm deviations boost turnout by 1.2 percentage points, whereas in the hottest regions, the same deviations reduce turnout by 0.73 percentage points. Temperature also affects electoral competition, with warm deviations increasing victory margins by nearly 1 percentage point. As climate change accelerates, these findings reveal how environmental factors may reshape democratic participation, with implications for electoral representation and policy outcomes in the world's largest democracy.

Keywords: Voter turnout, temperature, elections, climate change, political participation, gender gap, India, environmental shocks

1 Introduction

Severe weather events are becoming increasingly common as climate change accelerates, with growing consequences for economic decisions and human welfare. A large literature documents the effects of extreme temperatures on agricultural incomes (Blakeslee and Fishman, 2018; Burgess et al., 2014), human capital accumulation (Shah and Steinberg, 2017; Garg et al., 2020b), conflict and crime (Burke et al., 2015; Colmer, 2021), workplace safety (Park et al., 2021), and structural transformation (Liu et al., 2023). These impacts often fall disproportionately on vulnerable populations, including women, children, and poorer households (IPCC, 2022). Yet despite growing interest in the political economy of climate change, we know relatively little about how temperature shocks affect core democratic behaviors such as voter turnout, especially outside advanced democracies.

Understanding how heat affects electoral participation is increasingly important because climate conditions directly shape the costs of political participation. Elections require large-scale physical participation by voters, poll workers, candidates, and election officials—often under fixed schedules and with limited scope for short-run adaptation. During India’s 2024 general elections, multiple reports documented politicians and media personnel fainting due to heat exposure,¹ while at least 33 polling staff reportedly died from heat stroke on a single election day in Uttar Pradesh.² As temperatures continue to rise, such incidents are likely to become more frequent unless electoral systems adapt. Whether and how extreme heat alters who turns out to vote, therefore, has first-order implications for political representation and electoral outcomes.

This paper investigates how polling-day temperature affects voter turnout and electoral outcomes in Indian state assembly elections. India provides a useful setting for

¹<https://www.aljazeera.com/news/2024/5/14/india-election-will-a-forty-five-degree-heat-wave-affect-who-voters-choose>.

²<https://www.aljazeera.com/news/2024/6/2/heatstroke-kills-33-polling-staff-in-a-state-on-last-day-of-india-election>.

this analysis. As the world’s largest democracy, with over 900 million eligible voters, elections occur year-round across diverse climatic zones— from Himalayan regions to tropical coastal areas— generating substantial variation in heat exposure on election days. Moreover, as a rapidly warming developing country, India offers critical insights into how climate stress may shape political participation in other vulnerable democracies.

We assemble a comprehensive dataset covering state assembly elections from 1977 to 2023, combining constituency-level electoral outcomes with high-resolution temperature data measured during polling hours (7:30–17:30). Our empirical strategy exploits plausibly exogenous variation in polling-day temperature deviations from long-run historical norms,³ controlling for constituency fixed effects as well as year, month, and calendar day-of-year fixed effects. This approach isolates the impact of unexpected temperature realizations from seasonal patterns, long-run trends, and time-invariant local characteristics.

This paper makes four main findings.

First, polling-day temperature has robust and strongly non-linear effects on voter turnout. Temperature deviations exceeding 2°C above historical norms increase aggregate turnout by about 0.8 percentage points. These effects are asymmetric: warm deviations increase participation, while cold deviations have weaker and less consistent effects. To the best of our knowledge, existing work on election-day weather has largely focused on precipitation shocks or on developed-country settings (Gomez et al., 2007; Fujiwara et al., 2016; Van Assche et al., 2017). Emerging work in India studies how longer-run temperature exposure shapes electoral outcomes such as vote shares and political preferences (Amirapu et al., 2023). We study a distinct but complementary margin by focusing on unexpected temperature realizations on the day of voting and their effects on participation itself. Consistent with prior evidence (Van Assche et al., 2017; Amirapu et al., 2023), we

³We calculate location- and date-specific 30-year normal temperatures and measure deviations from these benchmarks on each polling day.

find that the temperature-voter turnout relationship is positive on average.

Second, we document significant gender heterogeneity in turnout responses to heat. Female voters respond more strongly to extreme warm deviations: deviations above 2°C increase female turnout by 0.83 percentage points, compared to 0.67 percentage points for men. As a result, higher temperatures modestly narrow the gender gap in turnout. These findings speak to a broader literature documenting persistent gender gaps in political participation, particularly in developing countries (Desposato and Norrander, 2009; Prillaman, 2023). They are consistent with the view that women’s political participation is especially sensitive to contextual constraints related to time use, mobility, and household responsibilities (Anukriti et al., 2020; Garg et al., 2020a).

Third, we show that the effects of heat on turnout vary sharply with long-run climatic conditions, revealing important limits to adaptation. In constituencies located in the coolest quartile of historical temperatures, warm deviations increase turnout by up to 1.5 percentage points. In contrast, in the hottest quartile, the same deviations *reduce* turnout by approximately 0.6 percentage points. This reversal suggests that moderate warming may facilitate participation in cooler regions, while populations already accustomed to high temperatures may face physiological or infrastructural thresholds beyond which additional heat suppresses turnout. This adaptation margin is largely absent from existing studies of weather and political behavior (Van Assche et al., 2017), which typically estimate average effects.

Fourth, temperature-induced changes in turnout translate into meaningful downstream electoral consequences. Victory margins increase by nearly 1 percentage point when polling-day temperatures exceed 2°C above normal, indicating that differential turnout responses across voter groups affect electoral competition. These effects are stronger in competitive elections and in unreserved constituencies, suggesting that environmental shocks interact with political institutions to shape representation. In doing so, we link

climate-induced participation shocks to downstream electoral outcomes, complementing work on climate and institutions that has largely focused on economic performance or conflict (Downs, 1957; Burke et al., 2015).

This paper advances three strands of scholarship. First, it extends research on weather and political participation, which is largely centered on precipitation shocks in advanced democracies (Gomez et al., 2007; Fujiwara et al., 2016; Van Assche et al., 2017), by documenting how extreme heat affects voter turnout in a major developing democracy. We show that temperature effects are strongly non-linear and can increase participation, challenging the view that adverse weather uniformly suppresses electoral engagement. Second, the paper contributes to the literature on gender gaps in political participation by demonstrating that environmental conditions can differentially affect male and female turnout, highlighting climate stress as an underexplored determinant of gendered political behavior (Desposato and Norrander, 2009; Prillaman, 2023). Third, it contributes to a growing literature on climate change and institutions by showing that temperature shocks affect not only economic outcomes but also the functioning of democratic processes themselves, with implications for political representation and electoral competition (Downs, 1957).

The remainder of this paper proceeds as follows. Section 2 describes India’s electoral institutions. Section 3 presents the data. Section 4 outlines the empirical strategy. Section 5 reports the main results, heterogeneity analysis, and mechanisms. Section 6 conducts robustness checks. Section 7 discusses the next steps. Section 8 concludes and discusses policy implications.

2 Institutional Context

Understanding how temperature affects turnout requires an institutional setting in which elections are frequent and salient, participation is primarily in-person on a fixed polling

day, and there is limited scope for strategic manipulation of election timing in response to short-run weather realizations. India's elections satisfy all three conditions and provide an opportunity to study how environmental shocks shape political participation in a large, competitive democracy. Below we briefly describe the structure of India's electoral system, the organization of election administration, and the system of reserved constituencies for historically marginalized groups.

2.1 Indian Electoral System

India is a federal parliamentary democracy in which executive authority at both the national and state levels is derived from legislative majorities. Our analysis focuses on elections to state legislative assemblies (*Vidhan Sabha*). These elections determine partisan control of state governments, which possess substantial authority over public goods provision, welfare programs, education, health, and local infrastructure. As a result, state assembly contests are typically high salience events for voters.

Each state is divided into single-member constituencies, and candidates compete under a first-past-the-post system. Assembly size ranges from 30 to 403 seats, reflecting population differences across states. Each state legislative assembly is in power formally for five years and has its own election schedule. Unlike contexts in which all constituencies vote on a single national election day, this staggered schedule generates both cross-sectional and temporal variation in polling-day temperatures.

The party system is fragmented and competitive. National parties, regional parties, and local entrants regularly contest state elections, and coalition governments are common. The effective number of parties is often well above two in many states. This competitive environment makes turnout consequential for electoral outcomes and creates incentives for parties to mobilize voters even under adverse conditions.

2.2 Electoral Administration

All national and state elections in India are administered by the Election Commission of India (ECI), a constitutionally independent body. It sets the electoral calendar, enforces the Model Code of Conduct, manages candidate and party registration, and oversees the deployment of security forces and polling personnel.

Elections for a given constituency occur on a single day with in-person voting within fixed hours, usually between 7:30 AM and 5:30 PM, with minor variation across states. India does not have absentee voting for the general population, and hence turnout is determined by whether eligible voters appear at their assigned polling station on the designated day.

Additionally, there is limited discretion to adjust election dates *ex post*. The ECI announces election schedules well in advance, often months before polling. While the Commission can take long-run climatic conditions into account when designing the overall election calendar (for example, by avoiding the peak monsoon in some regions), there is very little scope to reschedule specific constituencies in response to short-run weather forecasts. Once the schedule is announced, changes are rare and highly costly.

Voting infrastructure includes electronic voting machines (EVMs) at designated polling stations, most of which are public schools or other government buildings. ECI guidelines require that no voter should ordinarily have to travel more than a short distance from their residence to a polling station, leading to a dense network of stations in both urban and rural areas. However, the quality of physical infrastructure, such as the availability of shade, fans, or drinking water, varies across constituencies and over time. These differences create potential mechanisms through which temperature can affect participation, for instance, by increasing the cost of queuing at poorly equipped polling stations in already hot regions.

2.3 Reservation System

India’s Constitution mandates reserved legislative constituencies for Scheduled Castes (SCs) and Scheduled Tribes (STs), groups that have historically faced social and economic discrimination. In an SC-reserved (ST-reserved) constituency, only candidates belonging to SC (ST) communities may stand for election, while all eligible voters can vote. Approximately one-quarter of state assembly constituencies are reserved. This institutional feature has two implications for our analysis. First, it creates systematic variation in the social composition of the candidate pool and, potentially, in mobilization strategies, since parties may adjust campaign efforts when only SC/ST candidates can run. Second, reservation offers a natural dimension along which to examine whether temperature-induced changes in turnout differentially affect constituencies with larger shares of historically disadvantaged voters.

3 Data

3.1 Electoral Data

We assemble a new constituency-level dataset of state assembly elections from official records of the ECI. We study assembly elections in 32 unique states held between 1977 and 2023. The unit of analysis is the constituency-year, and the resulting panel covers 36,770 observations across major Indian states. For each election, we observe the full set of candidates, their parties, votes, and the size and composition of the registered electorate. This data is obtained from the Trivedi Centre for Political Data ([Trivedi Centre for Political Data, n.d.; Agarwal et al., 2021](#)). This structure allows us to link plausibly exogenous variation in temperature on polling day to rich measures of electoral participation and competition.

Our primary outcome is voter turnout, measured as total votes cast divided by the number of registered electors in the constituency, expressed as a percentage from 1 to 100.

Because the ECI reports votes separately by gender, we also construct male and female turnout rates using the corresponding gender-specific elector and vote totals. These measures allow us to examine whether heat shocks have heterogeneous effects on participation by gender, as well as map the effects of extreme heat on the gender gap in turnout.

In addition to turnout, we study several variables that capture the competitiveness and representational context of each election. Victory margin (measured as a percentage) is defined as the difference in vote share between the winning candidate and the runner-up, which provides a standard measure of how close the contest was. Electoral competition is summarized using a long-run constituency-level measure equal to the share of elections in our sample period in which more than three parties contested the seat; this proxy captures whether a constituency tends to experience fragmented, multi-party competition versus more consolidated two-party races. Reservation status indicates whether a constituency is reserved for Scheduled Caste (SC) or Scheduled Tribe (ST) candidates, as determined by the most recent delimitation.

Mean turnout in our sample is 68.7 percent of registered electors. Male turnout averages 70.5 percent, compared to 66.3 percent for women, implying a gender turnout gap of roughly 5 percentage points. The average victory margin is 13 percent, but there is substantial variation across constituencies, consistent with the presence of both safe and highly competitive seats.

3.2 Temperature Data

We utilize the ERA5-Land dataset to obtain weather information at a spatial resolution of 9 km grid spacing. This dataset, which is available from 1950 onwards, provides hourly information on surface variables such as daily maximum temperature and daily total precipitation for each constituency in our sample. For each constituency-election, we calculate the maximum temperature during polling hours (7:30-17:30) by first mapping ERA5-Land

grid cells to constituency polygons, then computing the area-weighted overlap between grids and constituencies, and finally calculating the weighted average temperature across overlapping cells.

To construct temperature deviations, we compute location and date-specific historical hourly means using 30-year data and take the average for the polling hours. This procedure captures the typical temperature for each location-date combination, allowing us to identify unusual temperature realizations.

3.3 Constituency Characteristics

We merge the electoral data with the Population Census of 1991, 2001, and 2011 using the Socioeconomic High-resolution Rural-Urban Geographic Dataset on India (SHRUG) data (*Census of India, 2011*; Asher et al., 2021).

3.4 Sample Characteristics

Figure 1 plots the density of elections across calendar months over our sample period. Elections are held almost every month of the year, with some seasonal clustering to avoid the rainy season. This reflects the staggered nature of India’s state electoral calendar.

Figure 2 presents a histogram of average polling-day temperature, which is approximately unimodal and centered on moderate temperatures, with relatively thinner tails at very low and very high values. Figure 3 and Figure 4 collapse these data at the level of the temperature deviation bins used in our empirical specifications. Most constituency-election observations occur under intermediate temperature conditions and within $\pm 2^\circ\text{C}$ of their historical normal, with meaningful mass in the more extreme bins. These descriptive patterns suggest that there is sufficient variation in both levels and unexpected deviations to identify the effects of heat on electoral behavior.

4 Empirical Strategy

4.1 Main Specifications

We employ two complementary approaches to estimate temperature effects on electoral outcomes.

4.1.1 Quadratic Specification

Our first approach models non-linear temperature effects using a quadratic function:

$$\text{Turnout}_{idt} = \beta_1 \cdot \text{Temp}_{idt} + \beta_2 \cdot \text{Temp}_{idt}^2 + \gamma \cdot 30\text{YrAvgTemp}_{idt} + \alpha_i + \delta_t + \mu_m(d) + \theta_{doy} + \varepsilon_{idt} \quad (1)$$

where Turnout_{idt} measures voter turnout in constituency i on date of polling d in year t ; Temp_{it} is average polling-day temperature; 30YrAvgTemp_{idt} is historical mean temperature for that location-date; α_i are constituency fixed effects⁴; δ_t are year fixed effects; $\mu_m(d)$ are month fixed effects; θ_{doy} are calendar day-of-year fixed effects. Standard errors are clustered at the constituency level.

4.1.2 Temperature Deviation Bins

To flexibly capture non-linearities, we also estimate:

$$\text{Turnout}_{idt} = \sum_k \beta_k \cdot \mathbb{I}[\text{TempDev}_{idt} \in \text{Bin}_k] + \gamma \cdot 30\text{YrAvgTemp}_{idt} + \alpha_i + \delta_t + \mu_m(d) + \theta_{doy} + \varepsilon_{idt} \quad (2)$$

where TempDev_{it} represents deviation between actual and historical temperature, categorized into bins: $< -2^\circ\text{C}$, -2 to -1°C , -1 to 0°C (omitted category), 0 to 1°C , 1 to 2°C ,

⁴Constituency fixed effects are defined at the pre- and post-delimitation level for each constituency.

$> 2^{\circ}\text{C}$.

Our identification exploits within-constituency variation in temperature deviations from historical norms. The rich fixed effects structure absorbs time-invariant constituency characteristics such as geography, infrastructure, and political culture through α_i , common temporal shocks such as national politics and economic conditions through δ_t , and seasonal patterns through μ_m and θ_{day} . Identifying variation therefore comes from unusual temperature realizations relative to what is typical for that specific location-date combination. For example, a constituency that normally experiences 25°C on March 15th but faces 30°C provides identifying variation.

4.2 Threats to Identification

The key identification assumption is that temperature deviations are orthogonal to unobserved determinants of turnout after conditioning on our fixed effects. Several factors support this assumption. First, short-term temperature fluctuations around historical norms are plausibly random with respect to political factors, providing plausible exogeneity. Second, our specification includes rich controls that account for location-specific, temporal, and seasonal factors. Finally, the sample is balanced since elections occur year-round across diverse locations, which limits concerns about systematic selection.

5 Results

5.1 Quadratic Temperature Effects

Table 1 presents results from the quadratic specification. Column (1) shows temperature has significant non-linear effects on aggregate turnout. The positive linear coefficient (0.303, $p < 0.01$) and negative quadratic coefficient (-0.003 , $p < 0.10$) confirm an inverted U-shape relationship. Turnout maximizes at 53.0°C , well above observed temperatures, im-

plying positive marginal effects throughout our sample range.

At mean temperature (26.8°C), a 1°C increase in polling day temperature increases turnout by 0.14 percentage points. This effect is economically meaningful—a one standard deviation temperature increase (4.5°C) increases turnout by 0.6 percentage points, comparable to get-out-the-vote campaigns.

Columns (2)-(3) examine gender-specific turnout. Male turnout shows stronger quadratic effects, maximizing at 44.0°C. Female turnout exhibits weaker non-linearity, with maximum at 114.9°C (effectively linear in our range). Column (4) reveals temperature significantly affects the gender gap—higher temperatures reduce male-female turnout differentials.

5.2 Temperature Deviation Effects

Table 2 reports the impact of heat on various measures of electoral outcomes, focusing on electoral turnout and the margin of victory. The effects on aggregate turnout increase monotonically with positive deviations from the average 30-year temperature (a graphic representation of these results is presented in Figure 5). A polling day with a temperature deviation exceeding 2°C is associated with an 1.16% (0.799 pp) increase in aggregate turnout and an increase of 7.65% (0.994 pp) in the victory margin, compared to polling days where the deviation falls within the reference temperature bin. A larger victory margin also suggests that elections held on hotter days are less competitive.

While positive temperature deviations from the reference temperature deviation bin positively affect turnout, the impact is 0.7-2.8 times higher for extreme temperatures. Interestingly, we also observe that female turnout rises by 1.25%, compared to a 0.95% increase for male turnout; however, the gender gap in turnout is not statistically significant at the 10% level of significance.

Our results are broadly consistent with existing work on temperature and elec-

toral participation. The estimated effects we document are smaller (half) than the average impacts reported by [Amirapu et al. \(2023\)](#) for India, which is likely attributable to key differences in scope and identification. While [Amirapu et al. \(2023\)](#) focus on election-year temperature exposure over the 2008–2017 period, we examine unexpected temperature realizations on the day of polling across a substantially longer time horizon. By isolating short-run polling-day shocks rather than cumulative exposure, our estimates capture a distinct participation margin.

In contrast, our polling-day temperature effects on turnout are substantially larger (~ 10 times) than those reported by [Van Assche et al. \(2017\)](#) in the United States. This difference is consistent with higher baseline exposure, lower adaptive capacity, and greater sensitivity to extreme heat in developing-country contexts. Taken together, our estimates fall well within the range of effects documented in the literature, while highlighting that extreme heat on election days is considerably more likely to shape democratic participation and electoral outcomes in climate-vulnerable democracies.

Benchmarking magnitudes against turnout interventions, our estimated effect of extreme polling-day heat (0.799 percentage points) is meaningful but not unprecedented. [Banerjee et al. \(2011\)](#) show that providing voters with candidate report cards increased turnout by about two percentage points, which is larger than our estimate. [Green et al. \(2012\)](#) summarizes a wide set of field experiments and reports weighted average impacts of 0.162 percentage points for direct mail, 4.8 percentage points for canvassing, and 0.98 percentage points for live phone calls. Relative to these benchmarks, our heat-induced turnout response is comparable in magnitude to live-call mobilization and significantly larger than the average direct-mail effect, although smaller than resource-intensive canvassing.

A similar comparison holds for women’s participation. [Prillaman \(2023\)](#) finds that an intervention aimed at women-led self-help groups increased female turnout in Madhya

Pradesh state elections by 0.046 percentage points. By contrast, we estimate that polling-day temperature deviations exceeding 2°C raise female turnout by 0.834 percentage points. This highlights that short-run climatic shocks can shift women’s political participation by amounts comparable to or greater than targeted empowerment interventions.

5.3 Heterogeneous Effects

5.3.1 Long term climatic conditions

We examine heterogeneity based on constituencies’ climate by dividing the sample into quartiles of historical average temperature. Table 3 and Table 4 present the results for aggregate turnout and victory margin, respectively. Striking patterns emerge. In the coolest quartile (Q1), warm deviations strongly increase turnout by 1.61%. Effects remain positive but attenuate moving to warmer quartiles. In the hottest quartile (Q4), warm deviations reduce turnout by 1%.

Figure 6 visualizes the relationship between electoral outcomes and temperature deviations within the highest quartile. Elevated temperature deviations are associated with a decline in voter turnout across gender groups and in aggregate, while concurrently exhibiting a positive association with the victory margin. These findings diverge from the existing literature and underscore the importance of adaptation margins, while highlighting that adaptation has inherent limits.

5.3.2 Reservation status

In Table 5 we present level effects to compare the effects between reserved and unreserved constituencies. A constituency is defined as unreserved if it is categorized as ‘General’ and reserved otherwise. Temperature effects are found to be concentrated in unreserved constituencies, where deviations exceeding 2°C increase turnout by 1.0 percentage point, whereas reserved constituencies show no statistically significant responses. Several mech-

anisms could explain this heterogeneity. Reserved constituencies may experience different competitive dynamics, SC/ST populations may face distinct socioeconomic constraints on voting, and geographic clustering may place reserved constituencies in regions with particular climate adaptations.

5.3.3 Long term electoral competitiveness

In Table 6 we present level effects to compare the effects between competitive and non-competitive constituencies, measured using effective number of parties (ENOP). We define elections as competitive when $ENOP \geq 3$. Competitive elections show stronger temperature responses, particularly for positive deviations. Both competitive and non-competitive constituencies experience an increase in aggregate turnout as a consequence of high deviation from average polling date temperature. Victory margins, however, respond differently by competitiveness. In non-competitive elections, temperature increases margins substantially (1.65 percentage points for $>2^\circ\text{C}$). In competitive elections, effects are muted and statistically insignificant, suggesting temperature-induced turnout changes have smaller impacts when multiple viable candidates exist.

5.3.4 Rural status

Table 7 reports level effects to facilitate a comparison between rural and urban constituencies. For this classification, constituency data are matched to the corresponding Population Census. A constituency is designated as rural if the weighted average share of its villages classified as rural is at least 50%. The effects on electoral outcomes are more pronounced in rural constituencies. This pattern aligns with the findings of [Amirapu et al. \(2023\)](#), which document that higher temperatures are associated with increased candidate investments in agricultural technologies, such as irrigation. These investments, in turn, enhance rural voter turnout and candidate popularity.

5.4 Channels

In Table 8 we test the impacts of high temperature deviations on winner's characteristics. We do not observe any significant effect of the highest temperature deviation on the likelihood of an incumbent or a turncoat winning the election. However, we find that elevated temperature deviations increase the probability of a recontestant winning by 4.62%, which is statistically significant at the 10% level of significance.

6 Robustness Checks

6.1 Time Period Adjustments

6.1.1 Panel construction

As specified in the empirical framework, constituency fixed effects are defined at the pre- and post-delimitation level for each constituency. To assess robustness, we overlay the post-delimitation and pre-delimitation shapefiles to identify constituency matches. For each pair, we retain the best match, defined as the highest spatial overlap between the two shapefiles. We then restrict the sample to constituencies with at least some overlap, enabling the construction of a panel structure.

Table 9 reports results from re-estimating the specification using this panel identification, with fixed effects applied to subsets based on overlap thresholds: at least 90% (column (1)), 95% (column (2)), 99% (column (3)), and 100% (column (4)). Aggregate turnout estimates remain broadly consistent with our main findings. While results in column (3) exhibit some noise, rendering them statistically insignificant, the direction of the coefficients remains stable across specifications.

6.1.2 Dividing the data into time periods

Next, we partition the sample into pre- and post-delimitation periods and re-estimate the main specification. Results for the pre-delimitation period primarily drive the observed turnout effects (Table 10). Interestingly, turnout is not statistically significant at the 10% level in the post-delimitation period, diverging from the findings of [Amirapu et al. \(2023\)](#). In contrast, estimates for victory margins remain consistent with our overall results.

6.2 Falsification Tests

To assess the validity of our estimates and rule out spurious correlations, we conduct falsification tests using election-cycle-specific characteristics. Specifically, we regress outcomes from the previous election on current temperature deviations. As expected, we find no evidence that temperature deviations influence the previous election cycle’s log of registered voters or the number of candidates (Table 11). These results strengthen confidence in the validity of our main findings.

7 Next Steps

Building on the evidence presented in previous sections, several avenues remain for deepening our understanding of the mechanisms and strengthening the robustness of the findings. To build on weather related robustness checks, we are currently working on adding precipitation controls, using the heat index (which is the feels-like temperature), and controlling for the temperature in the week of polling to show that the results remain consistent.

Additional work is needed to disentangle the relative importance of the different mechanisms through which temperature affects political participation. Physiological and comfort effects appear central, but additional evidence could clarify how these interact

with time allocation patterns, gender roles, and infrastructural constraints. In particular, we intend to use hourly turnout data by gender in local elections to better identify intra-day behavioral responses to temperature shocks. Descriptive evidence suggests that men and women vote at different times, which might explain the larger effect sizes of temperature on women’s electoral participation.

Finally, to deepen understanding of adaptation margins, we are examining general election data to assess the impact of extended polling hours introduced in select constituencies during the 2024 election cycle (primarily Phase 4 constituencies) on electoral outcomes.

8 Conclusion and Policy Implications

Our findings underscore that climate shocks can fundamentally reshape electoral dynamics—both who votes and who wins. Unexpectedly hot election days tend to increase turnout while simultaneously widening victory margins, thereby reducing electoral competitiveness. However, adaptation to heat has clear limits: in regions already experiencing high temperatures, additional heat suppresses participation. These patterns highlight the complex interplay between climate variability and democratic processes, underscoring the need to incorporate environmental factors into the design of electoral systems to build resilience.

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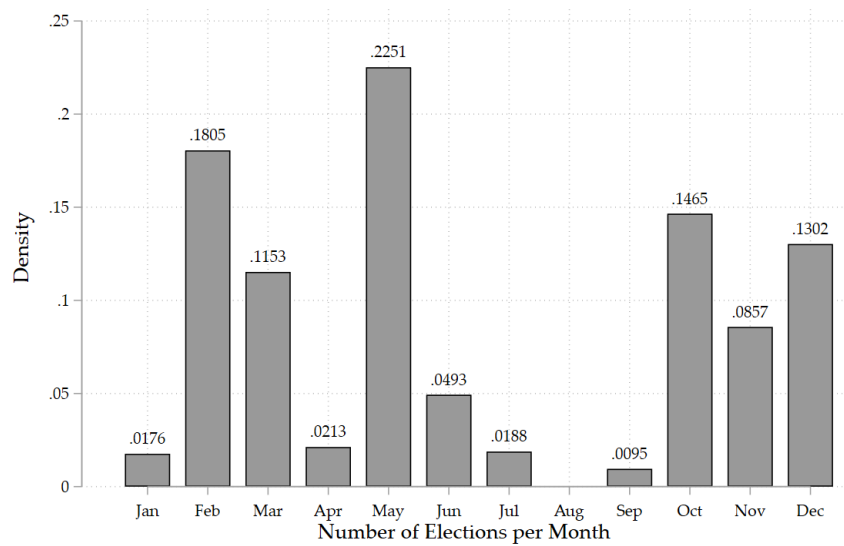
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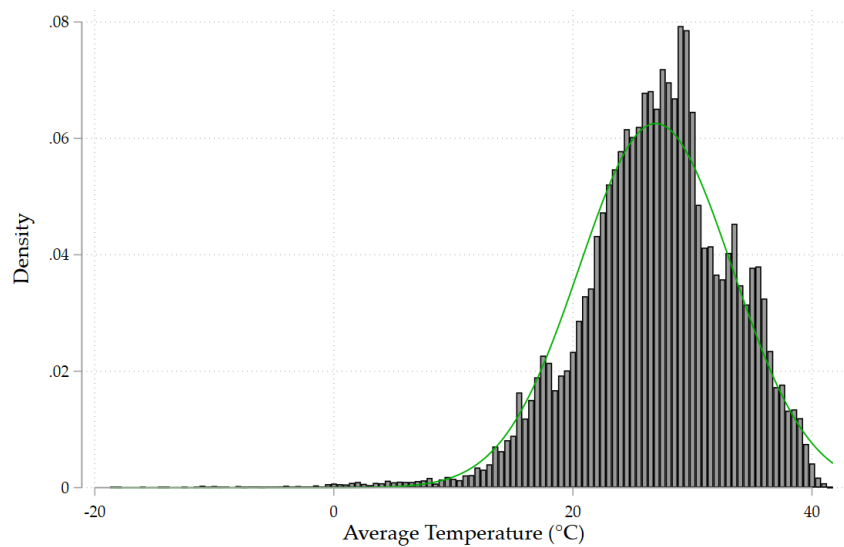
9 Figures

Figure 1: Density of Elections per Month in the Sample Period



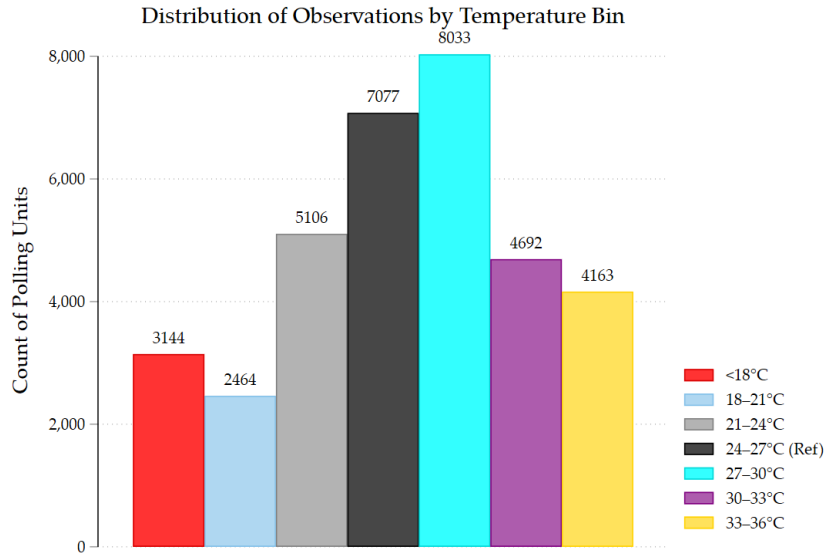
Notes: This figure plots the density of elections per month in the sample period of 1977-2023. The x-axis is the month in which polling took place and the y-axis plots the density.

Figure 2: Histogram of Average Temperature (0.5°C bins)



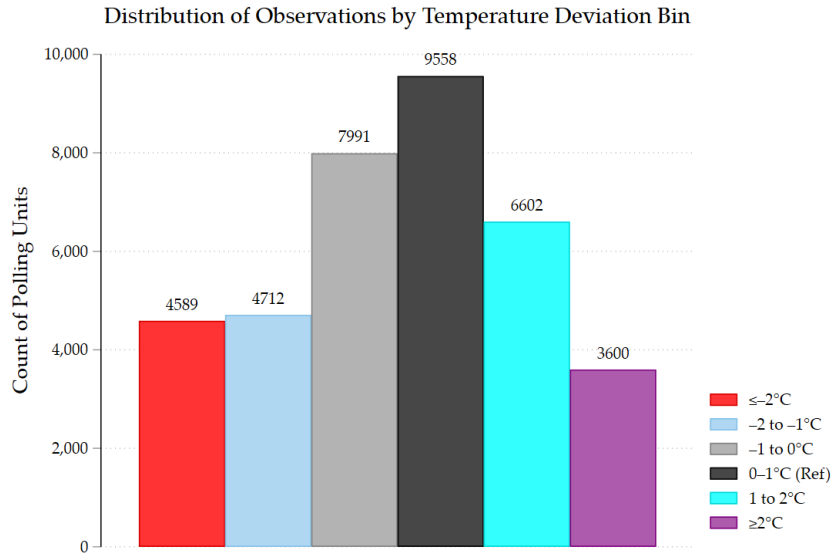
Notes: This figure plots the distribution of average daily temperatures in the sample period of 1977-2023. The histogram shows the frequency of observations across temperature ranges (in °C), with a bin width of 0.5°C. A normal distribution curve is overlaid to assess the degree of normality in the temperature data.

Figure 3: Distribution of Observations by Average Temperature Bin



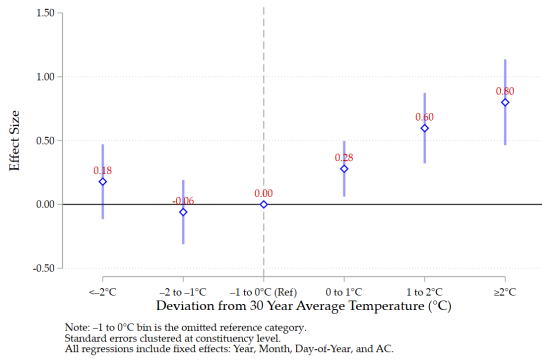
Notes: This figure plots the count of constituency-election cycles in each temperature bin for the sample period 1977-2023. Each bar represents a specific temperature range, and its height corresponds to the count within that range.

Figure 4: Distribution of Observations by Temperature Deviation Bin

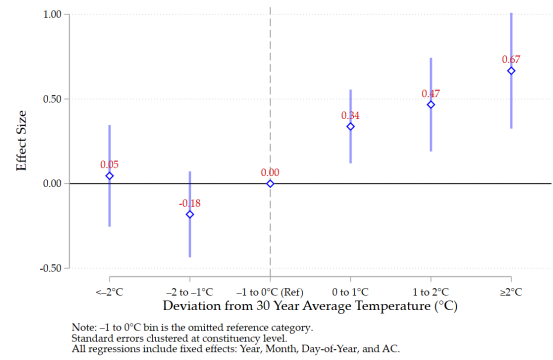


Notes: This figure plots the count of constituency-election cycles in the sample period of 1977-2023 by temperature deviation bin. The x-axis is the temperature deviation bin and the y-axis plots the count.

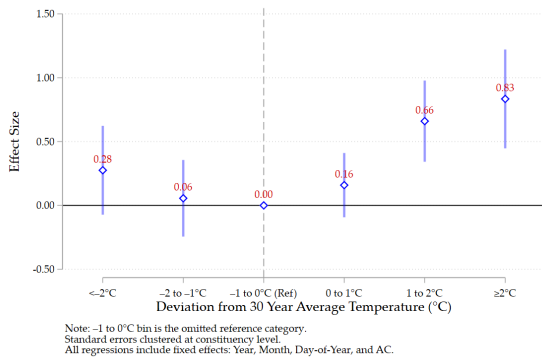
Figure 5: Effect of Polling Day Temperature Deviations on Turnout and Victory Margin



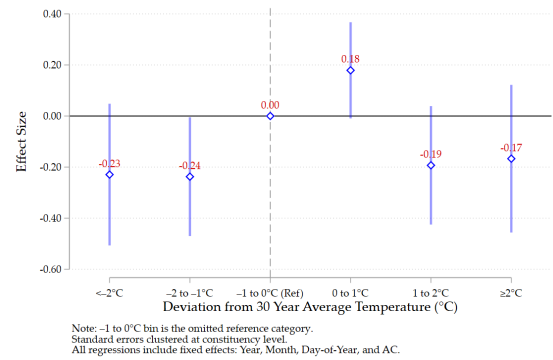
(a) Aggregate Turnout



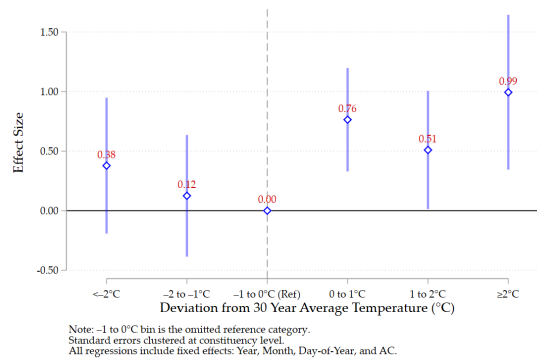
(b) Male Turnout



(c) Female Turnout



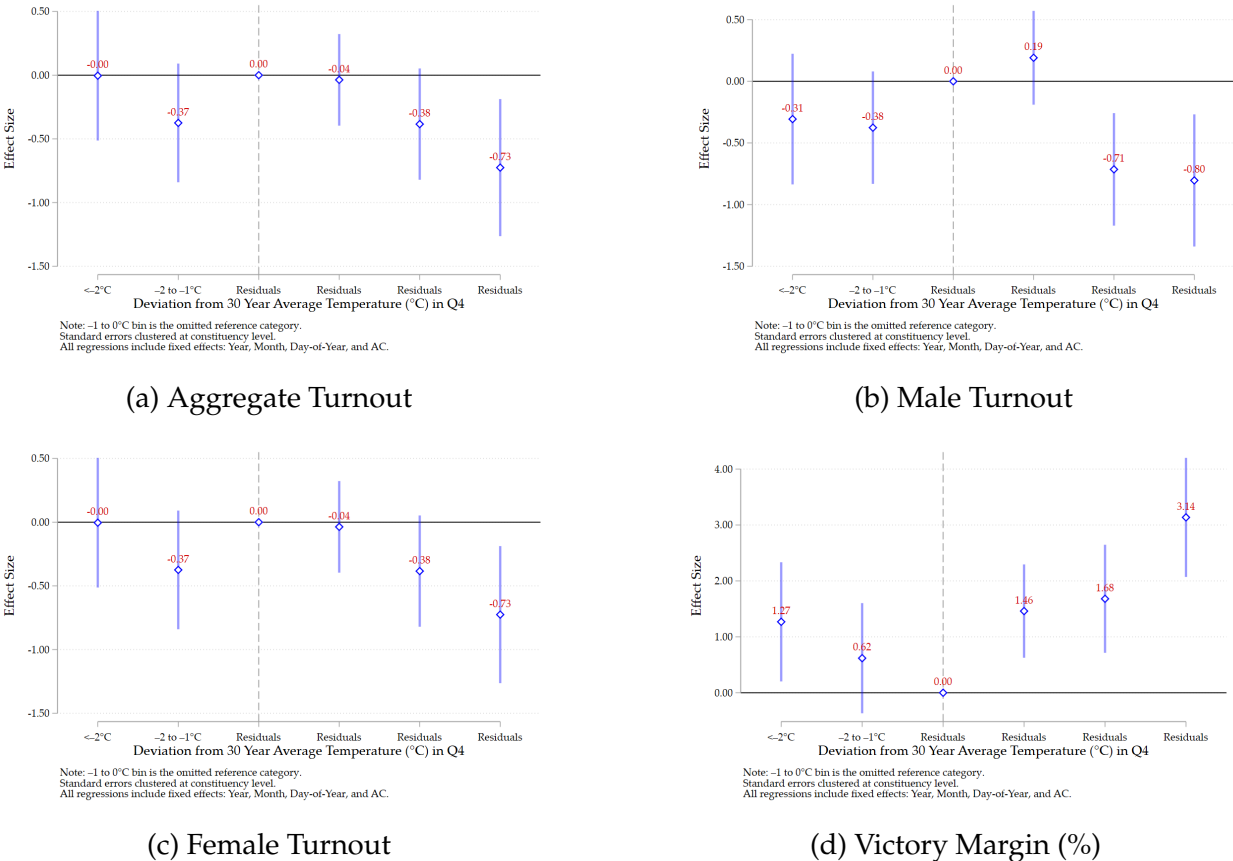
(d) Gender Gap in Turnout



(e) Victory Margin (%)

Notes: This figure shows the results from Table 2 where we run a fixed effects regression of the electoral outcomes on the whether the polling day's temperature deviation from the 30-year average lines in the specific deviation bin. The x-axis represents various temperature deviation bins, while the y-axis illustrates their respective coefficients, expressed as percentage point changes. These coefficients signify the effect on aggregate turnout in Panel (a), male turnout in Panel (b), female turnout in Panel (c), gender gap in turnout in Panel (d), and victory margin in Panel (e).

Figure 6: Effect of Polling Day Temperature Deviations on Turnout and Victory Margin in the Hottest Temperature Quartile



Notes: This figure shows the results for the constituencies which lie in the hottest regions (Q4). We run a fixed effects regression of the electoral outcomes on the whether the polling day's temperature deviation from the 30-year average lines in the specific deviation bin. The x-axis represents various temperature deviation bins, while the y-axis illustrates their respective coefficients, expressed as percentage point changes. These coefficients signify the effect on aggregate turnout in Panel (a) male turnout in Panel (b), female turnout in Panel (c), and victory margin in Panel (d).

10 Tables

Table 1: Quadratic Impact of Heat on Voter Turnout

	Aggregate	Male	Female	Gender Gap
	(1)	(2)	(3)	(4)
Avg. Temperature (°C)	0.303*** (0.090)	0.356*** (0.089)	0.202* (0.104)	0.155** (0.068)
Avg. Temperature Squared	-0.003* (0.001)	-0.004*** (0.001)	-0.001 (0.002)	-0.003*** (0.001)
Avg. Normal Temp (°C)	-0.503*** (0.048)	-0.487*** (0.048)	-0.523*** (0.058)	0.036 (0.041)
Mean Outcome (at Avg Temp)	66.93	69.15	64.12	5.03
Turning Point (°C)	53.03	44.02	114.87	24.39
p-value: Quadratic Terms	0.00	0.00	0.00	0.02
Observations	36770	36770	36770	36770

Estimates are from linear regressions of voter turnout outcomes on a quadratic function of polling day temperature and average normal temperatures. Fixed effects for constituency, year, month, and day-of-year are included in all regressions. Mean Outcome is the predicted value of the dependent variable at the sample mean of polling day temperature. The Turning Point reports the temperature (in °C) at which the quadratic prediction reaches its maximum or minimum. The p-value tests the joint significance of the linear and squared temperature terms. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 2: Impact of Heat on Voter Turnout and Margin of Victory

	Turnout				Margin of Victory
	Aggregate	Male	Female	Gender Gap	
	(1)	(2)	(3)	(4)	(5)
T Deviation <-2C	0.178 (0.149)	0.046 (0.153)	0.275 (0.177)	-0.229 (0.141)	0.378 (0.291)
T Deviation -2C to -1C	-0.060 (0.128)	-0.182 (0.130)	0.056 (0.153)	-0.237** (0.119)	0.125 (0.260)
T Deviation 0C to 1C	0.279** (0.111)	0.338*** (0.111)	0.159 (0.128)	0.179* (0.096)	0.764*** (0.221)
T Deviation 1C to 2C	0.597*** (0.141)	0.467*** (0.141)	0.660*** (0.162)	-0.193 (0.118)	0.510** (0.253)
T Deviation >2C	0.799*** (0.172)	0.667*** (0.174)	0.834*** (0.197)	-0.167 (0.147)	0.994*** (0.331)
Mean Outcome	68.70	70.55	66.28	4.26	13.00
Fixed Effects:					
Constituency	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes	Yes	Yes
Observations	36770	36770	36770	36770	35280

Estimates are from linear regressions with fixed effects. Columns (1)-(4) report voter turnout as a percentage of registered voters, and column (5) reports the margin of victory (%). All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 3: Impact of Heat on Aggregate Turnout

	Temperature Quartile			
	Q1 (1)	Q2 (2)	Q3 (3)	Q4 (4)
T Deviation <-2C	-0.868 (0.544)	0.450 (0.275)	0.559** (0.243)	-0.004 (0.259)
T Deviation -2C to -1C	-0.901** (0.363)	0.353 (0.260)	0.236 (0.196)	-0.375 (0.237)
T Deviation 0C to 1C	-0.584* (0.346)	0.769*** (0.220)	0.520*** (0.173)	-0.037 (0.183)
T Deviation 1C to 2C	-0.088 (0.540)	1.428*** (0.274)	0.975*** (0.194)	-0.384* (0.223)
T Deviation >2C	1.203** (0.470)	1.361*** (0.396)	1.661*** (0.259)	-0.726*** (0.274)
Mean Outcome	74.72	64.22	65.55	72.14
Average T	19.42	26.31	28.78	31.60
Fixed Effects:				
Constituency	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes	Yes
Observations	7013	9831	11311	8615

Estimates are from linear regressions with fixed effects. The dependent variable is the total voter turnout in percentage. We assign each constituency to a quartile based on its weighted average temperature, where Q1 corresponds to the coolest regions and Q4 to the hottest. Each constituency's weighted average temperature is computed by averaging contemporaneous election-day heat exposure, weighted by the number of electors in that year. All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. To present level effects, outcome and treatment variables are demeaned using a common set of fixed effects across all subgroup regressions. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 4: Impact of Heat on Victory Margin (%)

	Temperature Quartile			
	Q1 (1)	Q2 (2)	Q3 (3)	Q4 (4)
T Deviation <-2C	-0.186 (0.815)	-0.251 (0.547)	0.373 (0.510)	1.268** (0.543)
T Deviation -2C to -1C	0.142 (0.646)	0.631 (0.479)	-0.712 (0.475)	0.618 (0.502)
T Deviation 0C to 1C	0.971* (0.511)	-0.087 (0.447)	0.867** (0.398)	1.460*** (0.425)
T Deviation 1C to 2C	0.586 (0.621)	-0.601 (0.485)	0.499 (0.448)	1.681*** (0.492)
T Deviation >2C	-0.146 (0.697)	-1.666** (0.714)	1.666** (0.691)	3.136*** (0.543)
Mean Outcome	11.77	13.80	13.22	13.00
Average T	19.42	26.31	28.78	31.60
Fixed Effects:				
Constituency	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes	Yes
Observations	6725	9511	10434	8610

Estimates are from linear regressions with fixed effects. The dependent variable is the margin of victory of the winner in percentage. We assign each constituency to a quartile based on its weighted average temperature, where Q1 corresponds to the coolest regions and Q4 to the hottest. Each constituency's weighted average temperature is computed by averaging contemporaneous election-day heat exposure, weighted by the number of electors in that year. All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. To present level effects, outcome and treatment variables are demeaned using a common set of fixed effects across all subgroup regressions. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 5: Impact of Heat on Voter Turnout and Margin of Victory by Reservation Status of Constituency

	Not Reserved Constituencies				Reserved Constituencies			
	Aggregate	Male	Female	Margin	Aggregate	Male	Female	Margin
	Turnout	Turnout	Turnout	of Victory	Turnout	Turnout	Turnout	of Victory
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T Deviation <-2C	0.417*** (0.156)	0.221 (0.163)	0.577*** (0.188)	0.267 (0.321)	-0.558 (0.362)	-0.501 (0.360)	-0.646 (0.420)	0.750 (0.639)
T Deviation -2C to -1C	-0.176 (0.142)	-0.327** (0.145)	-0.029 (0.173)	0.130 (0.301)	0.292 (0.278)	0.246 (0.276)	0.325 (0.316)	0.127 (0.505)
T Deviation 0C to 1C	0.388*** (0.117)	0.430*** (0.119)	0.259* (0.141)	0.856*** (0.249)	-0.020 (0.261)	0.082 (0.257)	-0.115 (0.283)	0.496 (0.465)
T Deviation 1C to 2C	0.738*** (0.144)	0.607*** (0.150)	0.769*** (0.170)	0.556* (0.288)	0.204 (0.345)	0.080 (0.328)	0.350 (0.387)	0.366 (0.506)
T Deviation >2C	1.004*** (0.188)	0.772*** (0.195)	1.123*** (0.220)	1.153*** (0.383)	0.230 (0.374)	0.369 (0.367)	0.036 (0.421)	0.516 (0.641)
Mean Outcome	68.52	70.28	66.25	12.81	69.21	71.28	66.39	13.53
Fixed Effects:								
Constituency	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	27218	27218	27218	26303	9552	9552	9552	8977

Estimates are from linear regressions with fixed effects. Columns (1)-(3) and (5)-(7) report turnout as a percentage of registered voters, and columns (4) and (8) report the margin of victory (%). Sample is divided by reservation status of the assembly constituency in a given election, coding General constituencies as Not Reserved and all others as Reserved. Each constituency's weighted average temperature is computed by averaging contemporaneous election-day heat exposure, weighted by the number of electors in that year. All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 6: Impact of Heat on Voter Turnout and Margin of Victory by Competitiveness of Constituency

	Not Competitive Constituencies				Competitive Constituencies			
	Aggregate Turnout	Male Turnout	Female Turnout	Margin of Victory	Aggregate Turnout	Male Turnout	Female Turnout	Margin of Victory
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T Deviation <-2C	0.211 (0.208)	0.092 (0.204)	0.274 (0.238)	0.390 (0.415)	0.131 (0.213)	-0.021 (0.229)	0.274 (0.263)	0.348 (0.390)
T Deviation -2C to -1C	-0.121 (0.173)	-0.133 (0.168)	-0.134 (0.203)	-0.354 (0.366)	-0.002 (0.189)	-0.255 (0.200)	0.273 (0.230)	0.684* (0.354)
T Deviation 0C to 1C	0.422*** (0.150)	0.403*** (0.146)	0.366** (0.168)	1.385*** (0.301)	0.074 (0.160)	0.240 (0.169)	-0.135 (0.197)	-0.121 (0.318)
T Deviation 1C to 2C	0.822*** (0.198)	0.616*** (0.196)	0.926*** (0.220)	0.883*** (0.343)	0.300 (0.193)	0.277 (0.200)	0.300 (0.236)	-0.085 (0.366)
T Deviation >2C	0.917*** (0.235)	0.691*** (0.229)	1.047*** (0.265)	1.651*** (0.439)	0.603** (0.246)	0.611** (0.265)	0.498* (0.292)	-0.008 (0.491)
Mean Outcome	71.02	72.48	68.98	14.23	65.21	67.64	62.23	11.12
Fixed Effects:								
Constituency	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20730	20730	20730	20003	16040	16040	16040	15277

Estimates are from linear regressions with fixed effects. Columns (1)-(3) and (5)-(7) report turnout as a percentage of registered voters, and columns (4) and (8) report the margin of victory (%). Sample is divided by competitiveness of the assembly constituency in a given election, defining an election as competitive if the effective number of parties is at least three. Each constituency's weighted average temperature is computed by averaging contemporaneous election-day heat exposure, weighted by the number of electors in that year. All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 7: Impact of Heat on Voter Turnout and Margin of Victory by Rural versus Urban Classification of Constituency

	Rural Constituencies				Urban Constituencies			
	Aggregate Turnout	Male Turnout	Female Turnout	Margin of Victory	Aggregate Turnout	Male Turnout	Female Turnout	Margin of Victory
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T Deviation <-2C	0.033 (0.156)	-0.089 (0.161)	0.114 (0.186)	0.476 (0.302)	1.676*** (0.400)	1.442*** (0.398)	1.915*** (0.489)	-0.735 (0.750)
T Deviation -2C to -1C	-0.037 (0.134)	-0.162 (0.136)	0.082 (0.160)	0.262 (0.270)	-0.121 (0.346)	-0.213 (0.333)	-0.020 (0.417)	-1.231 (0.759)
T Deviation 0C to 1C	0.210* (0.116)	0.275** (0.116)	0.086 (0.134)	0.719*** (0.230)	1.108*** (0.276)	1.110*** (0.276)	1.028*** (0.327)	1.199* (0.639)
T Deviation 1C to 2C	0.605*** (0.147)	0.508*** (0.147)	0.631*** (0.169)	0.495* (0.260)	0.466 (0.347)	-0.063 (0.357)	0.971** (0.397)	0.552 (0.707)
T Deviation >2C	0.858*** (0.179)	0.748*** (0.183)	0.865*** (0.205)	0.886** (0.346)	0.066 (0.478)	-0.321 (0.462)	0.416 (0.560)	2.195** (0.868)
Mean Outcome	68.79	70.61	66.38	13.10	67.85	69.99	65.40	11.99
Fixed Effects:								
Constituency	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	33855	33855	33855	32429	2915	2915	2915	2851

Estimates are from linear regressions with fixed effects. Columns (1)-(3) and (5)-(7) report turnout as a percentage of registered voters, and columns (4) and (8) report the margin of victory (%). Sample is divided by the rural versus urban classification of the assembly constituency as per Population Census 1991, 2001 and 2011, classifying a constituency as rural if the weighted average share of its villages designated as rural is at least 50%. Each constituency's weighted average temperature is computed by averaging contemporaneous election-day heat exposure, weighted by the number of electors in that year. All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 8: Impact of Heat on Winner Characteristics

	Winner is Incumbent (1)	Winner Recontests (2)	Winner is Turncoat (3)
T Deviation <-2C	-0.012 (0.012)	0.012 (0.013)	-0.015* (0.008)
T Deviation -2C to -1C	-0.021** (0.010)	0.003 (0.011)	-0.015** (0.007)
T Deviation 0C to 1C	0.004 (0.009)	0.018* (0.009)	0.012** (0.006)
T Deviation 1C to 2C	0.023** (0.010)	0.031*** (0.011)	0.015** (0.007)
T Deviation >2C	0.011 (0.013)	0.024* (0.014)	0.010 (0.009)
Mean Outcome	0.33	0.52	0.11
Fixed Effects:			
Constituency	Yes	Yes	Yes
Year	Yes	Yes	Yes
Month	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes
Observations	35296	35296	35296

Estimates are from linear regressions with fixed effects. The dependent variables are the probabilities that, conditional on contesting, the incumbent wins (column (1)), whether the winner is a recontestant (column (2)), and whether the winner is a turncoat (column (3)). All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 9: Impact of Heat on Aggregate Turnout: Varying Definitions of a Panel

	$\geq 90\%$	$\geq 95\%$	$\geq 99\%$	100%
	(1)	(2)	(3)	(4)
T Deviation $< -2C$	-0.710 (0.529)	0.372 (0.751)	1.201 (1.735)	1.323 (3.057)
T Deviation $-2C$ to $-1C$	-0.804** (0.388)	-0.558 (0.591)	1.602 (1.605)	1.657 (1.153)
T Deviation $0C$ to $1C$	0.022 (0.352)	0.130 (0.491)	0.498 (1.154)	-0.749 (0.670)
T Deviation $1C$ to $2C$	0.673 (0.437)	0.856 (0.590)	1.296 (1.385)	2.281 (1.991)
T Deviation $> 2C$	0.859* (0.508)	1.137* (0.667)	1.624 (1.597)	4.175** (1.711)
Mean Outcome	67.03	66.73	66.52	66.53
Fixed Effects:				
Constituency	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes	Yes
Observations	4310	1745	366	108

Estimates are from linear regressions with fixed effects. Each column represents a different method for defining a panel, where the number indicates the minimum overlap between the pre- and post-delimitation shapefiles. The dependent variable is the total voter turnout in percentage. All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. Standard errors are clustered at the assembly constituency level.

* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$.

Table 10: Impact of Heat on Voter Turnout and Victory Margin by Time Period

	Pre-Delimitation			Post-Delimitation		
	Turnout (1)	Gender Gap (2)	Victory Margin (3)	Turnout (4)	Gender Gap (5)	Victory Margin (6)
T Deviation <-2C	0.026 (0.185)	-0.345* (0.182)	0.263 (0.361)	0.620*** (0.195)	0.574*** (0.124)	0.373 (0.551)
T Deviation -2C to -1C	0.146 (0.155)	-0.433*** (0.155)	-0.298 (0.317)	0.490*** (0.177)	0.069 (0.077)	1.266*** (0.441)
T Deviation 0C to 1C	0.244* (0.137)	0.270** (0.134)	0.575** (0.288)	0.396** (0.184)	0.067 (0.059)	1.199*** (0.335)
T Deviation 1C to 2C	0.434*** (0.168)	-0.124 (0.160)	0.502 (0.319)	0.406 (0.285)	-0.160* (0.082)	0.549 (0.435)
T Deviation >2C	0.406* (0.219)	0.142 (0.205)	0.968** (0.423)	0.417 (0.254)	-0.181 (0.112)	1.538*** (0.557)
Mean Outcome	66.84	6.91	13.90	71.79	-0.11	11.61
Fixed Effects:						
Constituency	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Month	Yes	Yes	Yes	Yes	Yes	Yes
Day of Year	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24776	24776	23302	11757	11757	11741

Estimates are from linear regressions with fixed effects. The dependent variables are the aggregate voter turnout in percentage, the gender gap in voter turnout in percentage, and the margin of victory of the winner in percentage. We present subgroup estimates by whether the election happened before or after the delimitation. All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. To present level effects, outcome and treatment variables are demeaned using a common set of fixed effects across all subgroup regressions. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.

Table 11: Impact of Heat on Electoral Outcomes in the Previous Election Cycle

	Log Total Registered Voters (1)	Number of Candidates (2)
T Deviation <-2C	0.017*** (0.002)	0.024 (0.161)
T Deviation -2C to -1C	0.000 (0.002)	-0.303*** (0.084)
T Deviation 0C to 1C	0.003** (0.002)	-0.015 (0.071)
T Deviation 1C to 2C	0.001 (0.002)	-0.239*** (0.086)
T Deviation >2C	0.001 (0.002)	-0.046 (0.111)
Mean Outcome	11.71	9.56
Fixed Effects:		
Constituency	Yes	Yes
Year	Yes	Yes
Month	Yes	Yes
Day of Year	Yes	Yes
Observations	36770	35296

Estimates are from linear regressions with fixed effects. The dependent variables represent outcomes from the current election cycle that occur before voting: natural log of the total number of registered voters and the total number of candidates contesting the election. All regressors are binary indicators denoting whether the contemporaneous temperature deviates from the constituency's historical average by an amount falling within each specified temperature bin. Standard errors are clustered at the assembly constituency level.

*p<0.1 **p<0.05 ***p<0.01.